NIHVA
NIHVA TECHNOLOGIES PVT. LTD.

Comprehensive AMC Of Telescopic Conveyor.

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1. General

a) Telescopic Conveyor was designed, developed, integrated by NIHVA

b) Various hardware and software components/subsystem of the Conveyor for which OEM's comprehensive maintenance required are detailed in Table-1. This specification details the technical requirements of Comprehensive Maintenance FO Telescopic Conveyor components/subsystem (Table 1) for a period of One year.

Table 1 Hardware and Software components/sub-systems to be covered under Comprehensive maintenance

Col 1	Col 2
Sr.No.	**Description**
I	**Mechanical Checks**
1	Roller
2	Chain Sprocket
3	Chain Assembly
4	Conveyor Belt
5	Mechanical Stopper
6	All hardware's
II	**Electrical Checks**
7	Control Panel
8	Interlocks
9	Sensors & Limit Switches
10	All Motors
III	**Hydraulic Items**
11	Power Pack
12	Coils
13	Hose Pipes
14	Hydraulic Cylinders

2. Scope of work

a) The scope includes preventive maintenance of the Mechanical and Electrical system (Table 1) where the satisfactory performance of the Conveyor shall be periodically verified following approved procedure and identification of defective components, parts or items. The identified defective/faulty items shall be replaced within the scope of contract.

b) The scope includes Breakdown maintenance of Conveyor (indicated in Table 1) in the event of a failure of the system (partly or total) and to bring the system back to service within response and resolution time frame indicated in **Col 3&4 of Table 3** which involves replacement of components, consumables and tools etc; the items not covered in Table 3 shall be brought to service within reasonable time

c) The replacement of faulty items shall be equivalent or better and the same shall be certified by NIHVA Technologies Pvt. Ltd Engineer-In-charge/project manager. If the defective components, parts or items are replaced, the defective part will be returned back to the contractor. Shifting and insurance charges of faulty spares, equipment and test instruments to theirworkshop shall be borne by the contractor.

d) Back to back agreement shall be entered or PO shall be placed on OEM/Authorized service providers for preventive maintenance and emergency visits for the following equipment.

Table 2 Maintenance frequencies for brought out items.

Sr No	Equipment	Frequency
1	Telescopic Conveyor	Quarterly

The agreement/PO with OEM/ Authorized service providers shall cover minimum visits by OEM representatives for carrying out preventive maintenance and Emergency visits in case of reported breakdown of system, however the contractor may decide on the additional visits if any required-on need basis. The systems/sub-systems within the scope of contract for which back-to-back agreement could not be entered, NIHVA Technologies Pvt. Ltd shall do the preventive maintenance of the system.

e) The contractor shall submit quarterly performance report covering all items within the scope of contract (Table 1) with preventive/breakdown maintenance report if any carried out during the quarter.

f) The complaints shall be attended and the reported problems/issues shall be rectified within the response and resolution period indicated in table 3 with standby equipment. The various spares in column 2 of table 3 shall be normalized with the period indicated in Column 4 of table 3 by taking out the standby equipment and thereby providing a permanent solution

Table 3 Response and Resolution requirements

Col 1	Col 2	Col 3	Col 4
Sr. No	**Description of item**	**Response and resolution Period(within)**	**Normalization by taking out standby equipment(within)**
1	Rollers	24hours (spare shall be maintained at site)	2weeks
2	Sensors & Limit Switch	24 hours (standby available)	2 weeks
3	Hose Pipes	72 hours	2 weeks
4	Chain	72 hours	2 weeks
5	Hydraulic Cylinder	120 hours	4 Weeks
6	Coils	72 hours	4 weeks
7	Power Pack	72 hours	2 weeks

g) Any disposal of faulty/defective items shall be as per NIHVA guidelines.

h) The details of material movement with gate pass No., status, etc., to be submitted to engineer-in-charge.

3. Technical requirements for CAMC of Hardware components

(Access controllers, UPS, Servers, computers, RFID printer etc. listed in Table 1, Section I and III)

NIHVA shall prepare and submit preventive maintenance schedule for all hardware items covered in the contract in consultation with OEM/OEM authorized dealer along with the bid. Among others the following are some of the hardware items for which the NIHVA shall submit preventive maintenance schedule for Engineer-in-charge's review and approval.

3.1 Preventive Maintenance of Telescopic Conveyor

i. The contractor shall depute qualified service engineers for planned periodic preventive maintenance activity of the system and attend breakdown maintenance as and when intimated by Engineer-in-charge/NIHVA representative.

ii. Approved procedure/manufacturer's instruction shall be followed while maintenance activities are carried out, thereby ensuring optimum performance and prevent any inadvertent degradation/damage to the system.

i. The service engineers shall follow site specific guidelines/safety instructions given by NIHVA engineers at site for fault rectification, preventive maintenance and healthinesscheck of all system components.

ii. The service engineer shall bring all the essential tools and tackles necessary to perform the planned job.

iii. The contractor/service engineer shall submit service report for each maintenance activity/breakdown call closure.

iv. The contractor shall arrange necessary transportation and accommodation requirements of his representatives at his own cost.

4. Availability of Spares at site

The contractor shall maintain requisite quantity of spares for all equipment covered under the contract to reduce system downtime, reduce delay in mobilizing components and thereby maximize overall system availability. The spare list in Table 4 is only indicative but not exhaustive.

Table 4 Minimum quantity of spare to be maintained at site

Sr No	Item	Qty
1	Rollers	2 (Nos)
2	Motors	2 Nos
3	Power Pack	2
4	Hydraulic Cylinder	2
5	Solenoids Coils	2 Set
6	Hose Pipe	2 Set

5. Documentation

The following documents shall be submitted to NPCIL by the contractor at different stages as listed below

a. Along with the bid

i. The bidder shall prepare and submit preventive maintenance schedule for all hardware items covered in the contract (Table 1) in consultation with OEM/OEM authorized service provider, the same shall form part of the contract on NPCIL acceptance.

b. Prior to the commencement of work

i. The AMC service provider shall prepare and submit preventive maintenance procedure complete with checklists/report template pertaining to all hardware systems covered in the scope, for NPCIL review and approval.

c. During the course of work

i. The AMC service provider shall submit quarterly performance report for all the items within the scope of contract along with preventive/breakdown maintenance report if any carried out during the quarter.

d. Along with the invoice for payment

i. The AMC service provider shall furnish a certificate stating that maintenance service for that particular quarter is completed satisfactorily, complete with endorsement and signature of Project manager/NPCIL authorized representative.

6. Commercial Offer for Comprehensive AMC

Hardware and Software components/sub-systems to be covered underComprehensive maintenance.

Col 1	Col 2	Col 3	Col 4
Sr.No.	**Description**	**Qty.**	**Price**
I	**Mechanical Checks**		₹ 13,50,000
1	Roller	4	
2	Chain Sprocket	8	
3	Chain Assembly	2	
4	Conveyor Belt	1	
5	Mechanical Stopper	2	
6	All hardware's		
II	**Electrical Checks**		
7	Control Panel	1	
8	Interlocks	3	
9	Sensors & Limit Switches	5	
10	All Motors	2	
III	**Hydraulic Items**		
11	Power Pack	1	
12	Coils	4	
13	Hose Pipes	3	
14	Hydraulic Cylinders	2	
		Total	***₹ 13,50,000***
		GST @ 18%	Extra at Actual
		P&F @ 3%	Extra at Actual
		Insurance Charges	Extra at Actual
		Unloading Chagres @ site	In Client Scope